

Puppet or Ansible: Choosing the Right Configuration Management Tool

Puppet and Ansible are both open source configuration management tools, each having its own strengths and weaknesses. This article will point the reader in the right direction when it comes to choosing one over the other.



uppet is basically a tool that helps you to manage and automate the configuration of servers.

It is one of the biggest names in the configuration management market. Developed around 2005 by an Oregon based software development company in Ruby, it is open source and runs on all major OSs like Linux, Windows, UNIX, macOS, etc.

Puppet allows us to define the state of our IT infrastructure, and then automatically enforces the correct state. It does not matter whether we are managing a few servers or thousands of physical or virtual machines, as it automates tasks that systems administrators do manually, thereby freeing up the time for them to work on the projects that deliver greater business value. Companies like Oracle and Google run their data servers with Puppet.

Ansible is an open source IT engine that automates application operation, cloud provisioning, intra service orchestration and other IT tools. It was

introduced in 2012 by AnsibleWorks, now owned by Red Hat.

As Ansible is developed in Python, it is lightweight and has fast deployment. It can also be run from CLI commands without using configuration files such as reboot, whether the server is running or not. For relatively complex tasks, its configuration is handled by the YAML syntax in a configuration file called a playbook. Ansible commands can be written in any programming language and be sent in the JSON format. It has also recently started supporting Windows.

Features common to both Puppet and Ansible

Before discussing the differences, we will consider the features common to Puppet and Ansible. Both are designed to automate the work involved in taking a physical or virtual machine from a generic configuration to a point where it serves the given tasks. These tasks may include installing packages, managing services, creating and updating configuration files and

running commands. Both have the idea of idempotency, which is important for automation. It means that when any automation script runs correctly, it should always leave the system in a consistent state, no matter how many times it runs and no matter what else has changed in that system. If partially run, and then run again, it should run correctly to completion just as if it were run from the beginning.

Puppet or Ansible tools are written to be resource models of the desired state. This is preferable to writing regular shell scripts which is very difficult.

Both support a rich template language for configuration files. Puppet's template is based on Embedded Ruby, while the Ansible template is based on Jinja2. Both have support for variables, whether fed in from a common set of files or set dynamically. To understand better, let us see a snippet of sample code of both – a simple example of putting a Hadoop configuration file into place.

The code snippet for Puppet is:

```
file { ['/opt/hadoop/etc/hadoop/hdfs-site.xml']:

    content => template('hadoop/hdfs-site.xml'),
    owner   => 'hadoop',
    group   => 'hadoop',
    mode    => '0644',
}
```

